PRD v2.0 — Week 13 Update

Property Name: Lava Birds B&B | **Island:** Hawai'i Island | **Visitor Segment:** Honeymooners

Overview of Three-Page App: The Lava Birds application consists of three main pages: a public marketing page, a public visitor dashboard, and a protected admin dashboard.

Tagline: A quiet, immersive stay near Kilauea, meant to be shared.

User Problem Statement: I am a budget-conscious traveler looking to book accommodations for my honeymoon. I can't find anything affordable online that offers unique and romantic experiences while allowing me to weigh my travel options. This is causing me stress and uncertainty.

The Opportunity

Hawai'i Island has a few existing B&B options near Kilauea, most of which offer somewhat affordable accommodations. Existing B&Bs, however, do not tailor guest experiences to reflect their proximity to Kilauea. Few offerings strongly target honeymooning couples, and packages that exist are usually found at luxury resorts. This fillable gap offers the unique opportunity to target a niche visitor segment: honeymooners seeking a romantic, immersive, and affordable experience.

Goals

The goal of the Lava Birds B&B application development is to create a responsive, easily navigable application for guests and system users. The application will aim to drive user interest, provide real-time data to help potential guests make informed decisions about travel dates, and provide an authentication-protected interface for system users.

Marketing Page

Marketing Page Requirements

Component	What is Displayed	Data Source	Interactive Behavior/ Responsive CSS Approach
Hero	Property name, tagline, island, main image/background video.	MongoDB via Express API	Hero CTA button that navigates to the booking contact form, Amenities button.
Special Offer Section	Current special offer and a CTA button prompting contact.	MongoDB via Express API	Button that opens a mailto link.
Image Carousel	Carousel of property images.	MongoDB via Express API	Navigable arrows.
About	Property description, location, target audience, lava.mp4, and owner information.	MongoDB via Express API	No major user interaction. Hover effects.
Amenities	Array of amenities: comfort-focused rooms, hot tubs, private outdoor access, local tea and coffee tastings, etc.	MongoDB via Express API	Renders amenities, no direct interaction.
CTA	Booking prompt, button, and a highlight of special offers.	MongoDB with some static fallback, as we are not developing a booking page at this stage.	Button that opens a mailto link.

Header/ Footer	Navigation links (Home, Dashboard, general nav, Admin) and copyright	Hardcoded and MongoDB/ Express API	Interactive navigation links that route users to the corresponding section.

User Personas

User Persona 1: Honeymooners

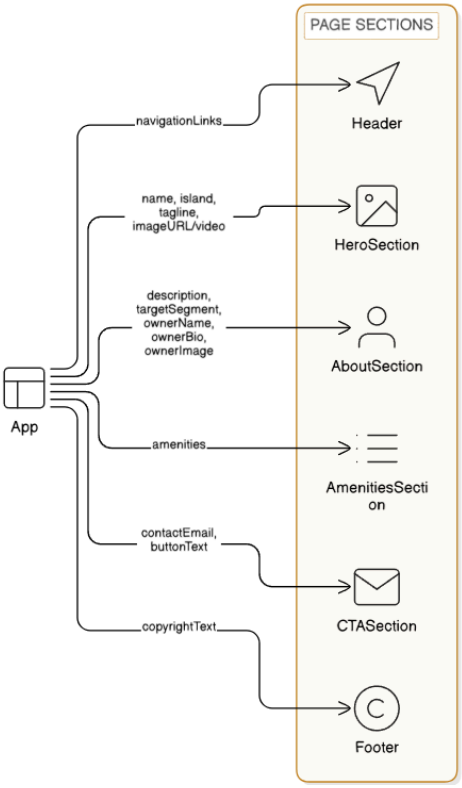
Profile/Role: A young couple who are planning a romantic honeymoon trip on a moderate budget.
Name: Maya and Ken
Goals:
- Find an affordable but memorable honeymoon experience
- Stay somewhere that feels unique and highlights the natural environment.
- Easily understand what the experience includes
- Make informed travel decisions

Pain Points:
- High costs of both unique and traditional luxury honeymoon experiences.
- Difficulty finding experiences that feel tailored to them without luxury pricing
- Potential for unpredictability in weather, volcanic activity, etc.

User Persona 2: Property Admin

Profile/Role: A non-technical property owner managing Lava Birds B&B content.
Name: Thomas
Goals:
- Easily update property information, amenities, and packages
- Post special offers and updates
- Keep the website data current without needing to update it daily
- Drive user interest

Pain Points:
- Limited technical knowledge
- Limited time to update the website daily
- Risk of outdated or inaccurate information



Original Component Hierarchy Diagram

Marketing Page Updates:
Week 13 [Weekly Edits]

Local Express server is now running with a functional fetch() call. React functional components and JSX are used throughout, with each major section added as its own component file. Updates to the database were made to tailor data and fields to Lava Birds: tagline, amenities, owner information, display images, logo, and a package offering. Alt attributes added to images comply with WCAG 2.1 AA standards. Added CORS to allow for cross-origin requests. Added a verified property ID to the code and confirmed database updates in MongoDB Compass. Designed the aesthetics and branding, ensuring a responsive layout.

Dashboard

Dashboard Design

	Arrivalchart	Originchart	Metriccards
Data Source	DBEDT visitor arrivals data	DBEDT visitor origin data	DBEDT and estimates
Chart Type	Bar Chart	Doughnut Chart	Three KPI cards
Why It Matters	Users can make informed decisions about when they'd like to arrive based on monthly trends.	Users can understand who visits Hawai'i Island.	Show travelers market information so they can make informed travel decisions based on trends.

Island Selector Ux

UI Control Chosen State	Chosen State Variable Name	Components That Re-Render When Selection Changes
Dropdown menu	slctIsland	Arrivalchart, Originchart, Metriccards, WeatherWidget

Weather Widget

	Placement On Page	City Queried	Loading State Handling; Data Fields Displayed
Weather	Top right of the page, under the top user interactive section	Hilo, Hawai‘i	“Loading” message, error handling; city name, temperature, conditions
Volcano (Potential to add livestream video beside it/ as a bg)	Top right of the page, to the right of the weather widget	Hilo, Hawai‘i (Kilauea)	“Loading” message, error handling; eruption data, conditions

Data Architecture

DBEDT Fields Used	JSON Schema Sketch	Express Route That Serves The Data
Island, Month, Arrivals, Metrics	{“ island”: "Hawai'i", “month”: "Feb", “arrivals”: 182000, “market” (by mo): {“japan”: 20000, “canada”: 8000, “other”: 9000 }, “metrics”: {“lengthofstay”: 8.9, “expenditure”: 250, “occupancy”: 73%}}	GET /api/arrivals?island=Hawai’i

Looking Forward To Weeks 14 And 15

Authentication Integration Plan

Implement admin authentication using Passport.js (local strategy) or JWT and create an admin login page. Store hashed passwords in MongoDB using bcrypt, implement CSRF protection, and add an admin-only feature allowing postings of special offers or announcements.

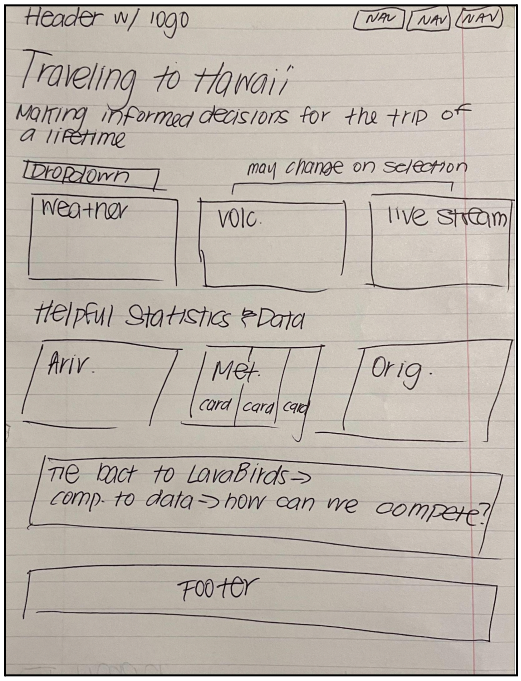
Open Design Questions

How will I refine the general aesthetic design? Do I still need to transition any hardcoded data to MongoDB? How will I implement authentication protocols? Where will announcements be posted, in what design format? How do I design the island selector? What additional islands should be included in the weather widget? Is the site easy to navigate? Does the site resolve both the problem statement and user pain points? What is the predicted conversion rate?

Important Acceptance Criteria for All Pages

Marketing Page:

- **Given** that a user visits the marketing page, **when** the page loads, **then** the lava video plays automatically without substantial load time and loops without interruption.
- **Given** that a user visits the marketing page, **when** they navigate around the page, **then** they book a stay at a 4% conversion rate.
- **Given** that a user visits the marketing page, **when** they navigate around the page, **then** they are able to easily identify amenities and booking processes within 20 seconds without confusion.



Dashboard Wireframe First-Draft

Dashboard:

- **Given** that a user visits the dashboard page, **when** the weather widget displays, **then** it loads within 1-2 seconds and displays the loading message while loading.
- **Given** that a user changes their island selection, **when** their choice is input, **then** Arrivalchart, Originchart, Metriccards, and WeatherWidget update within 1-2 seconds.
- **Given** that a user visits the dashboard page, **when** they would like to navigate to a specific section, **then** they are able to quickly and easily identify how to do so within 20 seconds without frustration, getting meaningful information from their search.